

Compact Dual-Band 3D MIMO Antenna for Wireless Biomedical Implantable Devices

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Abstract—In this paper, a compact dual-band 3D multi-input-multi-output (MIMO) antenna is designed for capsule endoscopic and leadless pacemaker implantable biomedical devices (IMBDs). The proposed MIMO antenna works for both Wireless Medical Telemetry Service (WMTS) 1.3975 GHz and Industrial, Scientific, and Medical (ISM) 2.45 GHz frequency bands. The proposed MIMO antenna, comprising four individual antennas, has overall dimensions of $5.12 \times 5.12 \times 4.6 \text{ mm}^3$. The antenna is encapsulated in a realistic devices and simulated in both homogeneous and heterogeneous body phantoms including heart, stomach, large intestine, and small intestine. The proposed MIMO antenna fabricated and subsequently tested in minced pork. The fabricated MIMO antenna demonstrated good performance including reflection coefficient (-22 dB , -19 dB), gain (-28.17 dBi , -18.15 dBi), -10 dB bandwidth (390 MHz, 670 MHz), minimal coupling (-23 dB , -24 dB), and fractional bandwidth (27%, 26%) at 1.3975 and 2.45 GHz, respectively. Furthermore, the specific absorption rate (SAR) and MIMO channel parameters including envelope correlation coefficient (ECC), and diversity gain (DG) further confirm the usefulness of proposed 3D MIMO antenna for wireless IMBDs.

I. INTRODUCTION

The demand for wireless IMBDs has significantly increased due to their broad applications in health monitoring systems, including capsule endoscopy, cardiac pacemakers, retinal prostheses, and head implants [1-3]. These devices enable real-time monitoring and data transmission to external systems and cloud platforms, supporting accurate diagnosis. Despite recent advancements, challenges remain, such as antenna miniaturization, multiband operation, wide bandwidth, sufficient gain, SAR, biocompatibility, and high data rates. While many designs use single-input-single-output (SISO) systems for their simplicity and acceptable performance, they are not ideal for next-generation devices, especially those with high-resolution cameras requiring fast image and video transmission. MIMO technology presents a promising solution by improving communication reliability, channel capacity, and signal quality while reducing fading crucial for real-time monitoring and diagnosis. However, it also introduces challenges, such as strong coupling between elements, the need for proper matching, and miniaturization for compact implants. To address these, various MIMO antennas have been proposed [4-7], but most have a two-dimensional geometry, limiting signal coverage and radiation diversity. This is particularly problematic in dynamic environment like capsule endoscopy, where device orientation continuously changes in the GI tract of patient.

Unlike planar designs, 3D MIMO antennas are better suited for biomedical applications involving continuously movement and posture variations. The 3D configuration enables continuous communication from all directions, enhancing pattern diversity, signal quality, and data rates. These advantages make 3D MIMO more suitable for reliable, high-performance wireless communication in complex, dynamic body environments. Designing such a compact 3D MIMO antenna with sufficient isolation, channel capacity, and pattern diversity, while fitting into implantable devices, remains challenging.

To address this, we present a 3D implantable MIMO antenna for biomedical devices operating under continuous movement and varying body postures scenarios. It functions at 1.3975 GHz and 2.45 GHz for wireless power transfer and data telemetry, with a compact volume of 120.58 mm^3 . The antenna provides stable performance across the GI tract including the stomach, small and large intestines as well as the cardiac system. The proposed MIMO antenna fabricated, encapsulated in realistic IMDs, and tested in minced pork. In order to show how safe the presented MIMO antenna is for the human body, a SAR value has been calculated. Additionally, MIMO parameters such as ECC, and DG has also been calculated.

II. 3D MIMO ANTENNA DESIGN

Fig. 1(a) and (b) shows Ant-1, which has compact dimensions of $4.6 \times 4.6 \times 0.26 \text{ mm}^3$. The thickness of both the substrate and the superstrate is 0.13 mm. To enable radiation pattern diversity in four opposite directions, Ant-1 is integrated on faces of a cube, as shown in Fig. 1(c) and (d). The 3D MIMO structure measures $5.12 \times 5.12 \times 4.6 \text{ mm}^3$, resulting in a total volume of 120.58 mm^3 . To achieve miniaturization a material with a high dielectric constant, Roger RT/duroid 6010, is used. It has a relative permittivity of $\epsilon_r = 10.2$ and a loss tangent $\tan\delta = 0.0023$. To ensure biocompatibility and reduce any negative effects on human tissue, the antenna is embedded in capsule-shaped devices made of biocompatible ceramic alumina (Al_2O_3), which has a dielectric constant of $\epsilon_r = 9.8$ and a loss tangent $\tan\delta = 0.006$. The capsule and pacemaker devices have diameters of 11 mm, wall thicknesses of 0.25 mm, and lengths of 24.65 mm and 24.5 mm, respectively, as illustrated in Fig. 2(c) and (d). A 50Ω coaxial feed port is used to excite the antenna. During the initial simulation, the encapsulated MIMO antenna was positioned at the center of a homogeneous small intestine phantom, chosen

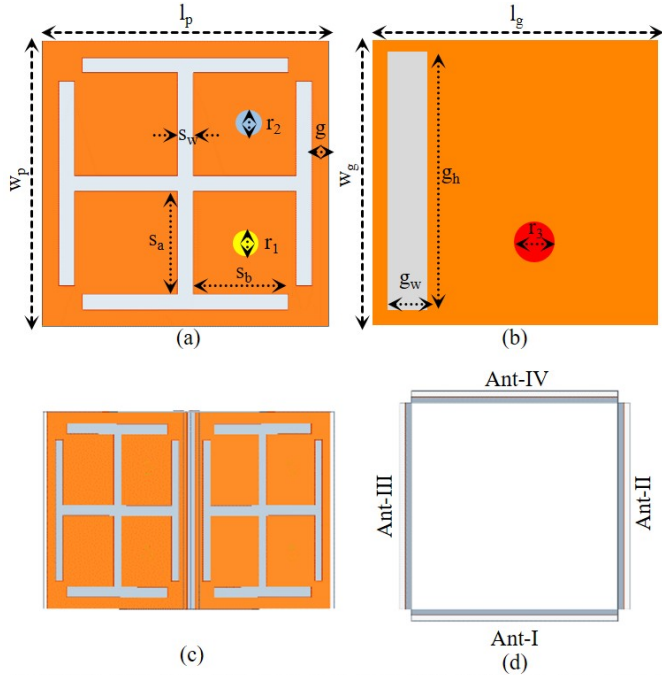


Fig. 1. (a) Ant-I patch (b) ground (c) cube (d) top view of cube ($l_p = w_p = 4.6$, $s_a = 1.8$, $s_b = 1.6$, $g = 0.27$, $h_1 = h_2 = 0.13$, $l_g = w_g = 4$, $g_w = 1$, $g_h = 3.5$, $r_1 = r_2 = 0.3$, and $r_3 = 0.6$).

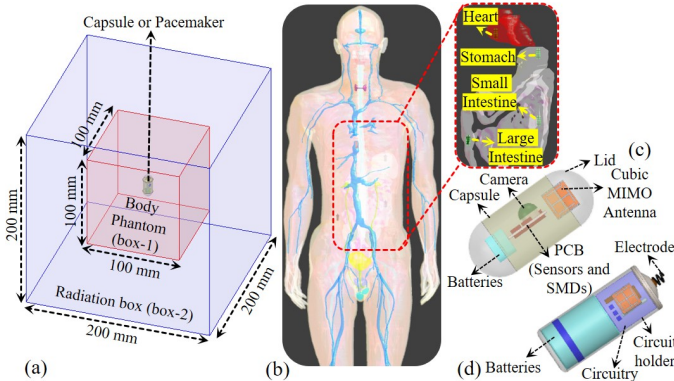


Fig. 2. (a) Homogeneous simulation setup (HFSS) (b) heterogeneous (Sim4Life) (c) capsule endoscopy device (d) cardiac leadless pacemaker.

due to its complexity within the GI tract. This phantom, labeled as box-1, has dimensions of 100 mm×100 mm×100 mm and is enclosed within a larger radiation boundary box (box-2) measuring 200 mm×200 mm×200 mm, as depicted in Fig. 2(a). The antenna was implanted at a depth of 50 mm. To more closely represent real tissue behavior, the dielectric properties of human tissue at 1.3975 GHz and 2.45 GHz were applied to box-1. The relative permittivity of the small intestine, large intestine, stomach and heart phantom measures 57.1, 56.1, 63.9, and 57.5 at 1.3975 GHz and 54.4, 53.9, 62.2, and 54.8 at 2.45 GHz respectively, while the electrical conductivity exhibits values of 2.44 S/m, 1.33 S/m, 1.44 S/m, and 1.51 S/m at 1.3975 GHz, and 3.17 S/m, 2.03 S/m, 2.21

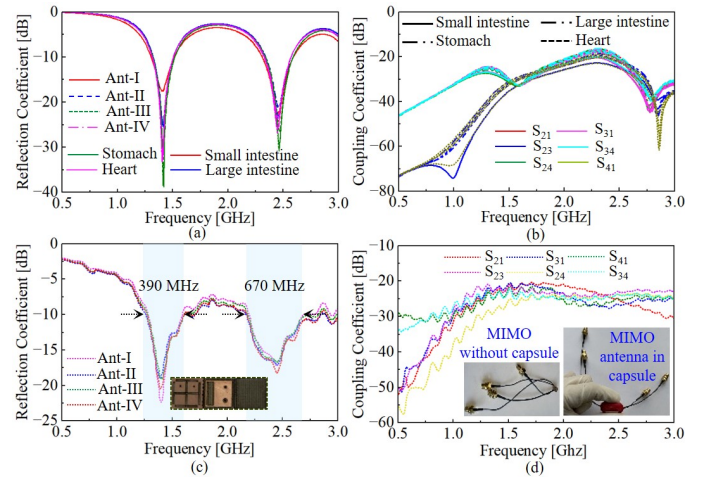


Fig. 3. Reflection and coupling coefficient (a,b) simulated (c, d) measured.

S/m, and 2.26 S/m at 2.45 GHz.

III. RESULTS AND DISCUSSION

The proposed cubic MIMO antenna demonstrates stable frequency performance and low mutual coupling, making it suitable multi tissue application such as the small intestine, large intestine, stomach, and heart. As shown in Fig. 3(a) and (b), the antenna consistently resonates within the intended frequency bands, with good reflection and coupling coefficients across all phantoms at both resonant frequencies. The antenna is also fabricated and enclosed in a 3D-printed capsule-like structure, as shown in the inset of Fig. 3(c) and (d). Scattering parameters are measured using a two-port vector network analyzer (VNA) with 50 Ω flexible subminiature-A (SMA) cables. During testing, two ports were connected to the VNA while the remaining ports were terminated with 50 Ω matched loads, and the procedure was repeated for all ports. As shown in Fig. 3(c), all four fabricated antenna elements $|S_{11}|$, $|S_{22}|$, $|S_{33}|$, and $|S_{44}|$ exhibited reflection coefficients below -22 dB, -18 dB, -19 dB, and -20 dB at 1.3975 GHz, and -18 dB, -18 dB, -17 dB, and -19 dB at 2.45 GHz. The antenna also achieved wide bandwidths of approximately 390 MHz and 670 MHz, with fractional bandwidths more than 27% and 26% at the lower and higher resonant frequencies, respectively. Simulated and measured radiation pattern of Ant-I at both frequencies, shown in Fig. 4, are in good agreement. The radiation patterns of other antenna elements (AntII-AntIV) are same but radiating in opposite directions to achieve radiation diversity. The measured gain is -28.17 dBi and -18.15 dBi at the lower and higher resonant frequencies, respectively.

To ensure the safety of the proposed MIMO antenna for human use, SAR values were calculated at both operating frequencies. Based on the IEEE C95.1-2005 standard, the maximum permissible SAR is 1.6 W/kg for 1-g tissue and 2 W/kg for 10-g tissue [8]. The antenna was placed in a heterogeneous environment, maintaining the same implant depth and positions as in the homogeneous setup. The simulation, performed using Sim4Life, is shown in Fig. 2(b). Each

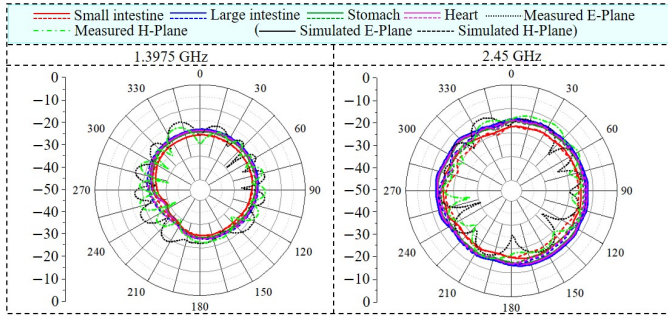


Fig. 4. Radiation pattern of Ant-I at both resonant frequencies.

antenna element was excited with 1 W input power. The calculated 10 g—SAR values and the corresponding maximum allowable input power (MAIP) are listed in Table I. Due to the input power limitation of -16 dBm for biomedical implants, the proposed antenna is confirmed to operate safely at both target frequencies. To calculate the degree of isolation or mutual correlation between antenna elements in a MIMO system we have calculated ECC. Although perfect isolation (zero correlation) is ideal, it is not practically achievable. Therefore, an ECC value below 0.5 is generally considered acceptable. In this work, ECC values were obtained from the far-field radiation characteristics of the proposed antenna. At the lower resonant frequency, the ECC values between ports 1, 2, 3, and 4 are 0.19, 0.11, and 0.17, while at the higher resonant frequency, they are 0.02, 0.08, and 0.04 as shown in Fig. 5(a). These low ECC values indicate strong isolation between elements, confirming the proposed MIMO antenna suitability for IMBDs that require reliable, high data-rate wireless communication. DG is another key parameter used to evaluate improvements in signal-to-interference ratio in MIMO systems. A DG value close to 10 dB typically indicates minimal correlation between antenna elements. For the proposed antenna, DG values at the lower resonant frequency between ports 1, 2, 3, and 4 are 9.2 dB, 9.3 dB, and 9.25 dB, respectively. At the higher frequency, the corresponding DG values are 9.8 dB, 9.5 dB, and 9.7 dB as shown in Fig. 5(b). All values exceed 9.2 dB across both frequency bands, reflecting good diversity performance and confirming the usefulness of antenna for biomedical applications.

IV. CONCLUSION

A compact dual-band 3D MIMO antenna has been developed for wireless IMBDs. It supports two operational frequencies including 1.3975 GHz and 2.45 GHz for wireless power transfer and data telemetry applications. The antenna features four elements arranged on a cube measuring $5.12 \times 5.12 \times 4.6$ mm³, resulting in a compact volume of 120.58 mm³. A prototype was fabricated and evaluated in a minced pork. Measured results include reflection coefficients of -22 dB and -19 dB, gains of -28.17 dBi and -18.15 dBi, -10 dB bandwidths of 390 MHz and 670 MHz, coupling coefficients of -23 dB and -24 dB, and fractional bandwidths of 27% and 26% at 1.3975 GHz and 2.45 GHz, respectively. SAR analysis

TABLE I
SAR AND THE CORRESPONDING MAIP FOR THE PROPOSED ANTENNA

Frequency	Maximum SAR (W/kg) and MAIP for 10-g (at 1 watt input power)			
	1.4 GHz		2.45 GHz	
	SAR (W/kg)	Input power (mW)	SAR (W/kg)	Input power (mW)
Small intestine	90.4	22.12	87.8	22.7
Large intestine	90.5	22.0	87.3	22.9
Stomach	90.1	22.1	87.2	22.9
Heart	90.5	22.0	89.4	22.3

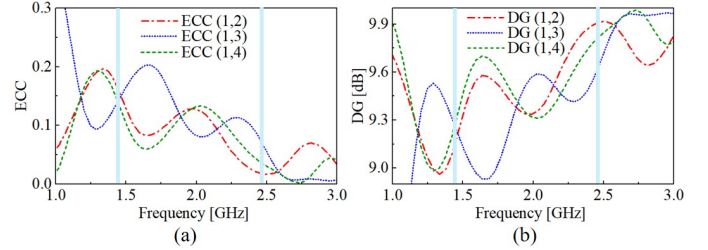


Fig. 5. MIMO channel parameters (a) ECC and (b) DG.

confirm that the proposed design is extremely safe for practical biomedical applications. MIMO performance parameters such as ECC and DG further confirm the antenna effectiveness.

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